

CUSC Appendix F4 — Protection Settings Schedule

Thornholme 400 kV connection to Creyke Beck

Connection point	Creyke Beck 400 kV
Voltage	400 kV
Scheme	Dual-redundant distance + backup
Issued by	Thornholme Renewables Ltd (protection engineer)

1. Protection philosophy

Main protection is provided by dual-redundant numerical distance relays on independent DC supplies and independent VT/CT cores. Backup is provided by directional earth-fault and overcurrent protection. Breaker-fail protection initiates busbar zone clearance.

2. Fault-clearance performance

The total fault-clearance time at the point of connection is 80 ms for a three-phase fault, within the NESO Creyke Beck zone requirement of 100 ms. The settings are co-ordinated with NESO's downstream zone-2 reach to maintain grading margins.

Schedule F4 — Relay settings

Function	Stage / zone	Setting	Time
Distance (main 1)	Zone 1	80% of line	instantaneous
Distance (main 1)	Zone 2	120% of line	350 ms
Distance (main 2)	Zone 1	80% of line	instantaneous
Directional E/F (backup)	—	0.2 In	500 ms
Overcurrent (backup)	—	1.2 In	800 ms
Breaker fail	—	stub	150 ms
Total clearance (3-ph)	—	—	80 ms